

Analysis of Hybrid Image Filtering Techniques

K.N.U.Ranasinghe

19/COM/311

EUSL/TC/IS/2019/COM/22

Submitted in partial fulfillment of the requirements for the award of the degree
Bachelor of Science in Computer Science [BSc (CS)] to the Department of Computer
Science, Faculty of Applied Science, Trincomalee Campus, Eastern University, Sri Lanka

Date of Submission

07/11/2024

Declaration

I hereby declare that the entire work embodied in this research work has been carried out by me. The extent of information derived from the existing literature has been documented and fully acknowledged at the appropriate places, the work is original and has not been submitted in part or full for any Diploma or Degree in this or any other University. I confirm that there is no plagiarism in this document and if detected, I abide by the action that will be taken for such plagiarism by the Faculty of Applied Science, Eastern University, Sri Lanka.

.....

K.N.U.Ranasinghe

19/COM/311

EUSL/TC/IS/2019/COM/22

Department of Computer Science

Faculty of Applied Science

Trincomalee Campus, Eastern University, Sri Lanka

Certification of the Supervisors

This is to certify that this research report entitled “**Analysis of Hybrid Image Filtering Techniques**” submitted by **K.N.U.Ranasinghe** for the degree of Bachelor of Science in Computer Science is a record of research work carried out by him/her under our guidance and direct supervision and that it has not been previously formed the basis for the award of any degree, diploma, associateship, fellowship or any other similar title.

This is also to certify the document represents the original independent work of the candidate.

.....

Signature of Co-Supervisor

Mrs. Thanushya Thanujan

Lecturer (Probationary)

Department of Computer Science

Trincomalee Campus, Eastern University, Sri Lanka

.....

Date

.....

Signature of Supervisor

Mrs. Kethika Krishnaraj

Lecturer (Confirmed)

Department of Computer Science

Trincomalee Campus, Eastern University, Sri Lanka

.....

Date

Acknowledgment

First and foremost, my heartily profound gratitude and appreciation are addressed to my co-supervisor **Mrs. Thanushya Thanujan** and the supervisor **Mrs. Kethika Krishnaraj** for their valuable supervision. His/Her advice, discussions and guidance were the real encouragement to complete this work. I admire his/her creativity, simplicity, generosity, work ethic, and ability to balance work and life. It has been an honour to work with him/her. I will always be thankful to him/her for the valuable time that they spent supervising my progress.

I would also like to thank **Mr.Subramaniam Thadchanamoorthy** - Head, Department of Computer Science, Faculty of Applied Science, Trincomalee Campus, Eastern University, Sri Lanka and all the lecturers of the faculty for facilitating and carrying out my research work.

Finally, I am indebted to my parents who have supported and encouraged me through their kindness and affection so that I could concentrate on my studies. They touched me more deeply than I could have ever expected.

Abstract

This project investigates hybrid image filtering techniques aimed at enhancing image quality through effective noise reduction, building on the strengths of traditional single-filter approaches. The study begins with the careful selection of a diverse dataset of high-resolution images that encompass various noise types, including Gaussian, salt-and-pepper, and speckle noise. By applying multiple filtering combinations, specifically focusing on Bilateral and Gaussian filters, the effectiveness of these techniques in both noise reduction and detail preservation is thoroughly evaluated. Key performance metrics Peak Signal-to-Noise Ratio (PSNR), Mean Squared Error (MSE), and Structural Similarity Index Measure (SSIM) are employed to quantitatively assess the quality of the filtered images. The analysis reveals that the Bilateral + Gaussian hybrid filter achieves the highest PSNR and the lowest MSE, demonstrating its superior ability to reduce noise while maintaining acceptable SSIM scores, which reflect strong visual fidelity. Furthermore, a comparative evaluation shows that individual filters, such as Bilateral and Non-Local Means, exhibit robust performance across different noise conditions. However, the hybrid approach significantly enhances the balance between effective noise suppression and the retention of fine image details, proving advantageous in various applications. This study underscores the potential of hybrid filtering techniques to produce clearer, more visually appealing images than conventional methods. The findings contribute valuable insights for professionals in image processing and related fields, advocating for the practical advantages of hybrid approaches in applications requiring high-quality image outputs. Ultimately, this research aims to encourage further exploration and application of hybrid filtering techniques in diverse image processing tasks.

Keywords: Hybrid Filtering Techniques, Noise Reduction, Image Quality Enhancement, Filtering Combination, Noise Types

Table of Contents

Chapter 01: Introduction.....	1
1.1 Project Overview.....	1
1.1.1 Aim of the Project.....	1
1.1.2 Audience	1
1.1.3 Scope.....	1
1.1.4 Approach.....	1
1.1.5 Assumptions	2
1.1.6 Expected Outcomes	2
Chapter 02: Related Work	3
2.1 Comparative Analysis of Noise Filtering Techniques in Digital Image Processing.....	3
2.2 Adaptive Orientation Filtering with Edge Preservation for Enhanced SNR	3
2.3 Hybrid Recommender System Using Collaborative and Content-Based Filtering.....	4
2.4 Adaptive Filtering Method for Mixed Noise Reduction in Images	4
2.5 Comparative Study of Hybrid Filter Topologies for DC Supply Distribution Systems	4
Chapter 03: Tools and Techniques	5
3.1 Tools.....	5
3.1.1 Python.....	5
3.1.2 OpenCV	5
3.1.3 NumPy.....	5
3.1.4 Scikit-Image	5
3.2 Techniques	6
3.2.1 Image Filtering Techniques.....	6
3.2.2 Hybrid Filtering Technique	6
3.2.3 Quality Metrics.....	6
3.2.6 Visualization and Comparison	6
Chapter 04: Methodology	7
4.1 Experiment Design.....	7
4.1.1 Dataset	7
4.1.2 Noise Simulation	7
4.1.3 Metric Evaluation	9
4.1.4 Selection of Best Filters	9

4.1.5 Development of Hybrid Filtering Technique	9
4.1.6 Analysis of Results	10
Chapter 05: Results and Discussion	11
5.1 Findings:.....	11
5.1.1 Individual Filter Results	11
5.1.2 Comparative Analysis:	11
5.2 Hybrid Filter Development	12
Chapter 06: Conclusion	14
Chapter 07: Future Work	15
7.1 Suggested Extensions.....	15
7.1.1 Real-Time Applications	15
7.1.2 Exploration of Adaptive and Machine Learning-Based Filtering	15
7.1.3 Application on Larger and Diverse Real-World Datasets	15
7.2 Limitations	15
7.2.1 Absence of Ground Truth Images:	15
7.2.2 Computational Demand of Hybrid Filtering:	15
References.....	16

Table of Figures

Figure 1: Sample Image From Dataset	7
Figure 2: Display Noise Simulation Example.....	8
Figure 3: Matrix for Filter Combination.....	9
Figure 4: Performance Metrics for Bilateral + Gaussian vs. Gaussian + Bilateral	10
Figure 5: Noisy Original image	10
Figure 6: After adding hybrid filter	10
Figure 7: PSNR value for all Combination	12
Figure 8: SSIM value for all Combination	13
Figure 9: MSE value for all Combination	13
Figure 10: Summary of Matrix	13

Chapter 01: Introduction

1.1 Project Overview

1.1.1 Aim of the Project

This project, titled "**Analysis of Hybrid Image Filtering Techniques**," is centered on developing a new hybrid filter by combining existing filtering methods for enhanced noise reduction and image quality. The study specifically investigates how the strengths of established filters, such as the Bilateral Filter and Non-Local Means Denoising, can be effectively integrated to improve performance beyond traditional filtering approaches. The primary goal is to achieve superior noise removal and detail preservation through innovative hybridization techniques.

1.1.2 Audience

This research is targeted at professionals and researchers in the field of image processing, especially those specializing in noise removal, denoising algorithms, and image quality enhancement. The findings from this study can potentially be applied in areas such as medical imaging, digital photography, and computer vision, where clarity and detail retention are critical.

1.1.3 Scope

The project will analyze the overall quality of images processed through different filtering techniques, focusing on key performance metrics: Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), and Mean Squared Error (MSE). By comparing the results of various filters based on these metrics, the study aims to identify which filter provides the best overall image quality, emphasizing effective noise reduction and detail preservation without isolating specific noise types.

1.1.4 Approach

The project utilizes Python, along with the OpenCV library, to implement the filtering methods. Each image will first be addressed by individual filters, such as Gaussian, Median, Bilateral, and Non-Local Means, with their performance meticulously documented. The hybrid filtering technique will then be developed and assessed. This comparative approach will include visual inspections of the filtered images and quantitative analysis based on PSNR, SSIM, and MSE, ensuring a comprehensive evaluation of each method.

1.1.5 Assumptions

Due to the lack of ground truth images, this study will rely on subjective evaluations and quantitative metrics to assess image quality. The combination of subjective assessments and metrics is expected to provide a well-rounded understanding of the hybrid technique's effectiveness compared to traditional methods.

1.1.6 Expected Outcomes

It is anticipated that the newly developed hybrid filtering technique will demonstrate superior results, indicated by higher PSNR and SSIM values and lower MSE scores. High PSNR and SSIM values are indicative of better image quality and noise reduction, while low MSE scores reflect reduced discrepancies between the original and processed images. These improvements will validate the hybrid method as a promising advancement for applications requiring detailed, high-quality images.

Chapter 02: Related Work

The field of image processing has made significant progress in noise reduction and image enhancement, though existing methods often face a trade-off between noise removal and detail preservation. This chapter reviews traditional and advanced filtering techniques, identifies their limitations, and positions the proposed hybrid approach as an innovative solution.

2.1 Comparative Analysis of Noise Filtering Techniques in Digital Image Processing

This research paper provides a comparative analysis of popular noise filtering techniques in digital image processing, including Median, Gaussian, Kuan, Morphological, Homomorphic, Bilateral, and Wavelet filters. Each filter's strengths are evaluated based on noise types like salt-and-pepper, speckle, and Gaussian. The study highlights the Wavelet filter's effectiveness in achieving the highest signal-to-noise ratio among these filters, making it suitable for diverse image noise conditions. This paper emphasizes the need for filter selection based on specific noise characteristics for optimal image quality. [1]

2.2 Adaptive Orientation Filtering with Edge Preservation for Enhanced SNR

This study introduces a novel filtering strategy that combines orientation-adaptive filtering with edge-preserving techniques, specifically designed to avoid cross-border filtering. The filter uses a fixed window size to adapt to local orientation by identifying high-anisotropy regions, leveraging generalized Kuwahara filtering for precise orientation detection. When compared to multi-scale filtering methods, this approach offers lower computational complexity and consistently improves Signal-to-Noise Ratio (SNR), demonstrating its effectiveness for high-quality image processing tasks. [2]

2.3 Hybrid Recommender System Using Collaborative and Content-Based Filtering

This paper addresses the challenges in content recommendation by proposing a movie recommender system that uses a hybrid approach, combining collaborative filtering with content-based filtering. The system leverages user preferences and movie attributes to recommend relevant movies for both new and existing users. By analyzing factors like popularity and attractiveness, this method offers enhanced accuracy and personalization in movie recommendations, supporting content providers in engaging users by aligning with their interests. [3]

2.4 Adaptive Filtering Method for Mixed Noise Reduction in Images

This paper proposes an adaptive filtering technique tailored to reduce mixed noise in images. Instead of relying on traditional methods like median or average filtering alone, which can be time-intensive and limited by noise type, the method combines adaptive masks and similarity calculations for each pixel. The adaptive mask design adjusts based on extreme noise values and applies weighted filtering, achieving effective removal of high-intensity impulse noise and reduction of mixed noise interference. [4]

2.5 Comparative Study of Hybrid Filter Topologies for DC Supply Distribution Systems

This paper evaluates two hybrid filter topologies one utilizing a single-leg inverter and the other employing an electronic smoothing inductor designed to mitigate current dynamics in DC supply systems, such as fuel cells. The active stages enhance low-frequency attenuation and manage surplus energy between the DC source and load. The findings reveal that the single-leg inverter filter provides superior attenuation, especially in low-frequency ranges, improving performance by nearly 15 dB compared to passive filters. [5]

Chapter 03: Tools and Techniques

This chapter provides a summary of the tools and techniques used to analyze the effectiveness of individual and hybrid image filtering methods.

3.1 Tools

3.1.1 Python

Python is the primary programming language utilized for its extensive libraries in image processing and data analysis.

3.1.2 OpenCV

OpenCV is used for loading images, applying filters, and simulating noise. Filtering techniques applied include:

- Gaussian Filter
- Median Filter
- Bilateral Filter
- Non-Local Means Denoising

3.1.3 NumPy

NumPy facilitates efficient numerical operations necessary for image manipulation and filtering.

3.1.4 Scikit-Image

Scikit-Image is employed for calculating image quality metrics such as:

- PSNR (Peak Signal-to-Noise Ratio)
- SSIM (Structural Similarity Index)
- MSE (Mean Squared Error)

3.2 Techniques

3.2.1 Image Filtering Techniques

The project examines various filtering algorithms, each evaluated for effectiveness against simulated noise.

- **Gaussian Filter:** A linear filter used to smooth the image and reduce Gaussian noise.
- **Median Filter:** A non-linear filter effective in reducing salt-and-pepper noise.
- **Bilateral Filter:** An edge-preserving filter that smooths the image while retaining important details, particularly at edges.
- **Non-Local Means Denoising:** An advanced denoising technique that reduces noise by comparing similar patches across the image, thus preserving more detail.

3.2.2 Hybrid Filtering Technique

The hybrid technique combines Bilateral and Non-Local Means filters to balance noise reduction and detail preservation.

3.2.3 Quality Metrics

Quality metrics (PSNR, SSIM, MSE) are calculated to quantify the performance of filtering techniques.

- **PSNR (Peak Signal-to-Noise Ratio):** Quantifies the image clarity after filtering, with higher values indicating less noise.
- **SSIM (Structural Similarity Index):** Measures the similarity between the original and filtered images, capturing structural fidelity.
- **MSE (Mean Squared Error):** Calculates the average squared difference between the original and filtered images, where lower values signify better noise reduction with minimal detail loss.

These metrics provide a standardized way to compare the effectiveness of each filter.

3.2.6 Visualization and Comparison

Results are visualized through metric plots for PSNR, SSIM, and MSE, enabling intuitive comparisons of filter performance.

Chapter 04: Methodology

4.1 Experiment Design

4.1.1 Dataset

A selection of high-resolution images from open-source libraries was curated for this study. These images cover a range of contexts, including landscapes and textured scenes, to provide a diverse basis for evaluating noise reduction techniques in different scenarios. [6]



Figure 1: Sample Image From Dataset

4.1.2 Noise Simulation

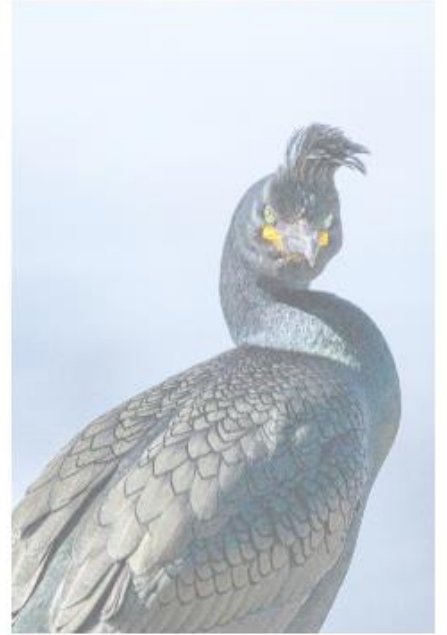
To emulate real-world noise conditions, three types of synthetic noise Gaussian, salt-and-pepper, and speckle were introduced to the images. This step was essential for assessing the filters' effectiveness in varied, controlled noisy environments:

- **Gaussian Noise** was added to simulate continuous, random brightness variations, common in natural images.
- **Salt-and-Pepper Noise** was used to mimic isolated bright and dark pixel disruptions, typically seen in transmission noise.
- **Speckle Noise** was introduced to represent noise caused by random brightness fluctuations, often observed in radar and ultrasound images

Original Image



Gaussian Noise



Salt and Pepper Noise



Speckle Noise



Figure 2: Display Noise Simulation Example

4.1.3 Metric Evaluation

Individual filters were initially evaluated based on PSNR, SSIM, and MSE metrics to assess each filter's ability to preserve image clarity and structural similarity. Following this, combinations of filters were analyzed, with these metrics calculated for each filter pairing. The resulting matrix of metrics revealed which filters complemented each other and which combinations delivered the best performance.

4.1.4 Selection of Best Filters

Through comparative analysis, Bilateral and Gaussian emerged as the most effective combination. This choice was supported by their superior performance across all metrics, with the Bilateral + Gaussian hybrid achieving the lowest MSE (210.99), highest PSNR (24.89), and highest SSIM (0.796) among the tested combinations. This indicated that the Bilateral + Gaussian combination effectively reduced noise while preserving image quality.

Matrix For Filter Combination			
	Non-Local + Median	Bilateral + Gaussian	Bilateral + Non- Local
PSNR	23.688579	24.888218	23.695477
SSIM	0.758132	0.796168	0.759205
MSE	278.113757	210.998379	277.672341

Figure 3: Matrix for Filter Combination

4.1.5 Development of Hybrid Filtering Technique

Using insights from the metric evaluation, a hybrid filtering technique was developed that combined Bilateral and Gaussian filters. This combination aimed to maximize noise reduction by leveraging Bilateral's strength in edge preservation and Gaussian's general smoothing ability. The resulting hybrid filter was designed to ensure robust performance across various noise types.

The effectiveness of the Bilateral-Gaussian hybrid filter was further supported by the following performance metrics:

- **PSNR:** 24.89 (Bilateral + Gaussian) vs. 24.08 (Gaussian + Bilateral)

- **SSIM:** 0.796 (Bilateral + Gaussian) vs. 0.768 (Gaussian + Bilateral)
- **MSE:** 210.99 (Bilateral + Gaussian) vs. 254.16 (Gaussian + Bilateral)

The results indicate that the combination of **Bilateral + Gaussian** filtering is the most suitable for noise reduction, outperforming the **Gaussian + Bilateral** arrangement. The Bilateral filter's edge-preserving qualities enhance the effectiveness of the Gaussian filter, resulting in superior image quality. This is evidenced by the metrics: **PSNR** of 24.888218 (Bilateral + Gaussian) versus 24.079705 (Gaussian + Bilateral), **SSIM** of 0.796168 compared to 0.76771, and **MSE** of 210.998379 against 254.16156. Thus, this hybrid approach effectively reduces noise while preserving structural similarity, so that the most suitable hybrid combination is **Bilateral + Gaussian**.

	Bilateral + Gaussian	Gaussian + Bilateral
PSNR	24.888218	24.079705
SSIM	0.796168	0.76771
MSE	210.998379	254.16156

Figure 4: Performance Metrics for Bilateral + Gaussian vs. Gaussian + Bilateral

4.1.6 Analysis of Results

The hybrid filter's performance was compared with that of individual filters, focusing on balancing efficiency and quality. The Bilateral + Gaussian hybrid demonstrated superior results across PSNR, SSIM, and MSE metrics, confirming its effectiveness in maintaining image detail while minimizing noise. This analysis guided the hybrid filter's design and provided a foundation for understanding the trade-offs involved in selecting filtering techniques for different noise conditions.



Figure 5: Noisy Original image

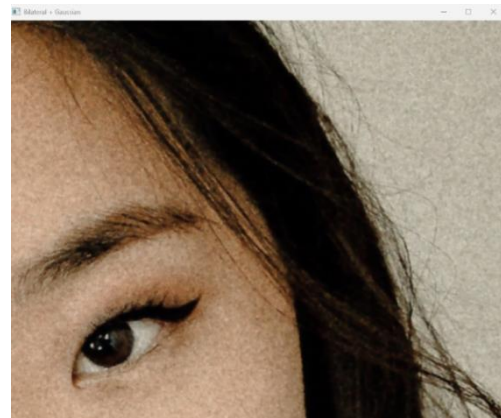


Figure 6: After adding hybrid filter

Chapter 05: Results and Discussion

5.1 Findings:

5.1.1 Individual Filter Results

The analysis begins by comparing the individual and combined performance of four filters Gaussian, Median, Bilateral, and Non-Local Means (Non-LM) using PSNR, SSIM, and MSE metrics to evaluate noise reduction quality. The following findings are based on the metrics observed in the tables below:

- **PSNR Comparison:** Higher PSNR values indicate better noise suppression. As shown in the "PSNR Values for All Combinations" table (Figure 7), the Bilateral filter achieves the highest PSNR (28.20) when used alone, demonstrating strong noise suppression. In combinations, Bilateral maintains high PSNR values, particularly in the Bilateral-Gaussian pairing, which exhibits a significant improvement in noise reduction
- **SSIM Comparison:** SSIM is a measure of structural similarity; higher SSIM reflects better preservation of image structure. As seen in the "SSIM Values for All Combinations" table (Figure 8), the combination of Bilateral and Gaussian filters achieves high SSIM values, indicating effective noise reduction while retaining important structural details.
- **MSE Comparison:** Lower MSE suggests less error in noise removal. According to the "MSE Values for All Combinations" table (Figure 9), the Bilateral filter, particularly in the Bilateral-Gaussian combination, achieves a significantly lower MSE, confirming its effectiveness in reducing error.
- **Summary of Metrics:** The "Summary of Metrics" table (Figure) demonstrates that the Bilateral-Gaussian hybrid filter generally performs best across PSNR, SSIM, and MSE metrics, making it an optimal candidate for noise reduction in various scenarios

5.1.2 Comparative Analysis:

Visual and Quantitative Comparison: The numerical data in these tables provides a clear basis for evaluating the effectiveness of various filter combinations. The Bilateral-Gaussian combination consistently shows high PSNR, high SSIM, and low MSE across different noise types, effectively balancing noise reduction with detail retention. This reinforces the choice of the Bilateral-Gaussian hybrid filter as a strong solution for noise suppression.

5.2 Hybrid Filter Development

Based on the findings, a hybrid filtering technique was developed by combining the Bilateral and Gaussian filters. This choice is supported by the individual strengths of each filter, as indicated by the metrics:

- **Filter Selection Rationale for Hybrid Approach:** The Bilateral and Gaussian filters were selected based on their distinct advantages. The Bilateral filter achieved high PSNR and SSIM values across combinations, demonstrating its strength in edge preservation and effective noise suppression. The Gaussian filter complements Bilateral by providing general smoothing across the image. Together, they form a balanced hybrid filter that maximizes noise reduction while preserving image quality.
- **Performance Evaluation:** The hybrid filter's performance was rigorously evaluated against individual filters. The combination of Bilateral and Gaussian filters demonstrated superior results across all key metrics (PSNR, SSIM, MSE), confirming its effectiveness in maintaining image detail while minimizing noise. This analysis informed the design of the hybrid filter and provided insights into the trade-offs involved in selecting filtering techniques for various noise conditions.

PSNR	gaussian	median	bilateral	non-LM
gaussian		25.448851	27.652514	26.76863
median			26.077267	25.420592
bilateral				27.291774
non-LM				

Figure 7: PSNR value for all Combination

SSIM	gaussian	median	bilateral	non-LM
gaussian		0.822074	0.884623	0.874714
median			0.840593	0.832395
bilateral				0.889518
non-LM				

Figure 8: SSIM value for all Combination

MSE	gaussian	median	bilateral	non-LM
gaussian		185.436534	111.642357	136.84136
median			160.455141	186.64706
bilateral				121.31177
non-LM				

Figure 9: MSE value for all Combination

	gaussian	median	bilateral	non-LM
PSNR	26.660015	25.277715	27.304959	26.322731
SSIM	0.861766	0.815913	0.876793	0.862533
MSE	142.679093	195.451542	122.966889	153.860719

Figure 10: Summary of Matrix

Chapter 06: Conclusion

This study aimed to develop a hybrid image filtering technique that effectively reduces noise while preserving image detail. Through a systematic analysis of existing filters Gaussian, Median, Bilateral, and Non-Local Means, each filter's effectiveness was evaluated using metrics such as PSNR, SSIM, and MSE.

The findings revealed that the combination of Bilateral and Gaussian filters emerged as the most effective hybrid approach. The Bilateral filter excels at preserving edges while reducing noise, and the Gaussian filter provides effective general smoothing. By leveraging the strengths of both filters, the Bilateral-Gaussian hybrid demonstrated superior performance in noise reduction and detail preservation compared to individual filters.

The results confirm that this hybrid filter not only enhances noise suppression but also maintains image clarity, making it suitable for applications requiring high-quality image restoration. This project contributes a valuable filtering technique that can be applied in fields where image quality is critical, such as medical imaging, remote sensing, and photography.

Future research could further optimize the Bilateral-Gaussian hybrid filter for real-time applications and assess its performance on larger, more diverse real-world datasets. Such efforts would enhance the practical robustness of the hybrid filtering technique and broaden its applicability across various domains.

Chapter 07: Future Work

7.1 Suggested Extensions

This research on developing an optimized hybrid image filtering technique highlights several directions for future exploration:

7.1.1 Real-Time Applications

To enable real-time noise reduction, further work could involve optimizing the hybrid filter through parallel processing or GPU acceleration, making it suitable for dynamic applications like live streaming, surveillance, or diagnostic imaging.

7.1.2 Exploration of Adaptive and Machine Learning-Based Filtering

Future studies could investigate combining the hybrid filter with adaptive machine learning models that adjust filter strength in real-time based on noise density or type. This dynamic approach could improve filter effectiveness in complex noise environments, enabling more selective and efficient noise reduction.

7.1.3 Application on Larger and Diverse Real-World Datasets

Testing the hybrid filter on diverse datasets, such as naturally noisy low-light or medical images, would further validate its robustness and adaptability. This could provide insights into its utility across different domains, particularly in handling varied real-world noise.

7.2 Limitations

While the results of this study show promising outcomes, certain limitations should be acknowledged:

7.2.1 Absence of Ground Truth Images:

The lack of ground truth images poses a challenge for precisely quantifying noise reduction. While synthetic noise provides a basis for evaluation, real-world noise often presents more complex patterns, which may impact filter effectiveness.

7.2.2 Computational Demand of Hybrid Filtering:

Although the hybrid filter improves noise reduction, the combined computational load may limit its application in real-time or resource-limited settings. While well-suited for post-processing, further refinement is needed to improve its efficiency in environments where quick processing is essential.

References

- [1] J. S. S. D. a. N. G. Sanjib Das1, "A COMPARATIVE STUDY OF DIFFERENT NOISE FILTERING TECHNIQUES IN DIGITAL IMAGES," *Engineering Research and General Science*, 2015.
- [2] P. Bakker, L. v. Vliet and P. Verbeek, "Edge preserving orientation adaptive filtering," *IEEE*, p. 52, 1999.
- [3] A. H. A. u. C. f. a. C. b. F. f. R. System, "G Geetha1, M Safa1, C Fancy1 and D Saranya2," *Conference Series*, 2018.
- [4] T.-J. Su and C.-I. Li, "An Adaptive Filtering Method for Mixed Noise of Images," *IEEE*, 2012.
- [5] S. Ostroznik, P. Bajec and P. Zajec, "A Study of a Hybrid Filter," *IEEE*, 2009.
- [6] "unsplash," [Online]. Available: <https://unsplash.com/>. [Accessed 30 10 2024].
- [7] G. I. a. R. Marudhachalam2, "NEW HYBRID FILTERING TECHNIQUES FOR REMOVAL OF GAUSSIAN NOISE FROM MEDICAL IMAGES," p. 5, 2011.
- [8] scikit-image.org, "Image filtering — Image analysis in Python," [Online]. Available: https://scikit-image.org/skimage-tutorials/lectures/1_image_filters.html. [Accessed 23 09 2024].
- [9] Bansal, "Denoising Images in Python - A Step-By-Step Guide - AskPython," 30 09 2021. [Online]. Available: <https://www.askpython.com/python/examples/denoising-images-in-python>. [Accessed 4 10 2024].
- [10] H. Vedoveli, "Image Filtering Techniques in Image Processing — Part 1," 25 8 2023. [Online]. Available: <https://medium.com/@henriquevedoveli/image-filtering-techniques-in-image-processing-part-1-d03362fc73b7>. [Accessed 7 10 2024].
- [11] Stanford.edu, "Stanford.edu," 2019. [Online]. Available: <https://ai.stanford.edu/~syyeung/cvweb/tutorial1.html>. [Accessed 8 10 2024].